

August 2025

Certification: Letter of Recommendation

It is my pleasure to certify and recommend **Md. Al Amin Chy** for successfully completing the second half (last and most important part) of the private online thesis course/program in the area: **Optics/Photonics/Electromagnetics** at [Mahdy Research Academy](#) under my supervision and the guidance of my team of Young Experts (YE). To the best of my knowledge, it is Bangladesh's first private research academy targeting high-quality private theses (an EdTech of advanced research targeting Q1 journal publication, which so far operates in an online setup).

This letter of recommendation is being issued after a thorough review of the final assignment: a complete draft of a journal article derived from the research project undertaken by **Md. Al Amin Chy** during the second half of the course. This phase comprised around 20 final sessions/online classes out of a total of approximately 40 (first half: more than 20 online sessions). Based on the depth, rigor, and quality of the submitted work/manuscript, I am confident that the article holds strong potential for publication in a reputable Q1 or Q2 journal.

*I have not certified anyone else whose identity is not given on the website below. All the participants are available in the list given on the website. Notably, this letter has been provided **only to the participants**:*

- (i) whose articles have already been submitted in a reputed journal by me or*
- (ii) whose draft version of the journal articles have been submitted to me, which will be submitted in a top journal after further polishing.*

List of participants in this part (second half) is given below and the name of **Md. Al Amin Chy** should be found in this list (*this is just the list of participants; only a few participants among them have received this letter*):

[Online Private Research Course/Program: List of all the participants for the final part](#)

Mahdy Research Academy's private online thesis and research program aims to democratize research, cultivate an impactful research community in Bangladesh, and guide participants toward producing publishable work in top-tier (Q1/Q2) journals. The first half of the program (January 2024 – June 2024) enrolled over 800 participants [**in 4 different areas: 1. Optics/Photonics 2. Quantum Mechanics 3. Quantum Computing/information 4. Applied AI: Machine & Deep learning**], while the second half (August 2024 – December 2024) saw over 500 enrollees [in those 4 areas], **including several junior faculty members, graduate students, and advanced undergraduates from leading Bangladeshi and international universities**. Each online session was held for around 2.5 to 3 hours.

Md. Al Amin Chy (1st author) has worked on a project titled, “A Multifunctional Metasurface Design: Achieving Superior Beam Steering, RCS Reduction, and Electromagnetic Wave Manipulation”

Their proposed metasurface employs a cost-effective FR-4 substrate as its unit cell, demonstrating exceptional electromagnetic wave reflection properties. Across the high-frequency spectrum of 50–100 GHz, the metasurface exhibits amplitude and phase responses comparable to those of a conventional copper plate, except at approximately 91 GHz, where it achieves a significant dip in reflection. Their metasurface efficiently helps to propagate surface waves while maintaining substantial electric field and magnetic field amplitudes of nearly 1 V/m and 5×10^{-3} A/m, respectively, contrasting starkly with a standard copper plate, which fails to sustain these fields over the entire surface. Moreover, it demonstrates noteworthy radiation control by redirecting incident energy into a predefined direction (67° in the phi plane) while simultaneously minimizing radiation across a full 360° angular spectrum. Compared to a copper plate, which exhibits an unstable RCS ranging from -20 dB to 0 dB, their metasurface maintains a significantly reduced and stable monostatic RCS of -40 to -30 dB across an ultra-broadband range of 50 GHz. Their findings hold immense potential for stealth technology, radar applications, and next-generation wireless communications.

They were a thesis group of 3 people, and one assigned young expert from Mahdy Research Academy also guided them carefully under my overall supervision. I hereby certify that Md. Al Amin Chy successfully completed the second half (last and most important part) of the online private thesis program and initially submitted a draft journal article based on their research. Later, the quality of their revised final article is of a standard suitable for publication in a Q1 or Q2 journal and has already been submitted in a top journal.

Dr. Mahdy Rahman Chowdhury
Professor, Dept of ECE, North South University
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The first half of this research course, conducted online in a private setting, covered an extensive syllabus ranging from foundational (i.e. basic to advanced electromagnetics/optics) to advanced topics in photonics/optics research, mathematical analysis and literature review. The curriculum of the first half included discussion (& detailed simulations) on several published journal articles, covering a range of topics such as: optical manipulation, tractor beams, metasurface based lateral force, solar sails, metamaterial absorbers, nonlinear optics: 3rd harmonic generation, photonic time crystal and photonic space-time crystal. These discussions and simulations also included my own published research. Several simulation tools and software have been demonstrated (e.g. COMSOL Multiphysics, CST Microwave Studio, Wolfram Mathematica etc.). We also held in-depth technical discussions on how to interpret and explain/analysis the results and figures of a journal article.

In the second half of the course, participants were introduced to a novel research idea, with the primary objective of applying the theoretical and practical knowledge gained in the first phase. They engaged in iterative cycles of trial and error to refine their methodologies and achieve meaningful results. Instruction during this phase emphasized analyzing prior research strategies, deepening conceptual understanding of relevant physical principles, and solving problems through established theoretical frameworks. Participants also received dedicated sessions on OriginLab for high quality data analysis and graph creation, Blender for crafting polished scientific figures and 3D visualizations, and Canva. To ensure depth and quality, ample time was dedicated to developing and submitting a complete draft of a research manuscript, accompanied by a professionally designed poster presentation summarizing their findings. Only those who completed the research draft paper (targeting a top journal article), poster, and video presentation have earned this Recommendation. **Md. Al Amin Chy** is one of them.

The final manuscript exemplifies the growth of **Md. Al Amin Chy** throughout the course. The results are well-structured and supported with high quality visuals, and the discussion thoughtfully places the findings in relation to existing literature, suggesting realistic paths for future work and publication. From my point of view, the manuscript meets the criteria expected for Q1/Q2 journals in terms of scientific depth, clarity, and novelty. Without the knowledge of these ideas and techniques (given below), it would not be possible to complete the journal article:

1. Each group was initially tasked with reproducing simulation/ code-based experimental results from a previously published journal article selected from a pool of Optics/Photonics related papers.
2. Strong command in electromagnetic/optical theory and relevant calculation techniques (connected with the project), demonstrating deep understanding of optical/photonic principles.
3. Notable simulation knowledge of either COMSOL OR CST MICROWAVE STUDIO [based on the assigned project]
4. Participants and their respective groups selected a specific novel idea of focus. Each group was assigned to a Young Expert (YE), who provided ongoing support and addressed the group's questions and challenges on a weekly basis. Several advanced relevant theories were discussed during the online live classes (both central & YE sessions).
5. How to write and wrap up a journal article. How to write a draft copy of the cover letter (to the editor of a journal publishing group).
6. I and my team also collected their video presentation regarding their own draft journal article along with a soft copy of the poster (a one-page poster describing their detailed work).

I believe that **Md. Al Amin Chy** demonstrated a strong commitment to learning, as evidenced by the decision to enroll in this intensive private thesis course. The completed version of the journal article further supports my **strong recommendation**. I am confident that **Md. Al Amin Chy** will excel in the topics covered during the second half (last and most important part) of the research program in the area: **Optics/Photonics/Electromagnetics** at Mahdy Research Academy. If you have any further questions, please feel free to contact me.

Dr. Mahdy Rahman Chowdhury.

Signature: 

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